Phenomenological Regimes Induced by Structural Identity

A Classification of Structurally Admissible Experience Types Under Identity Rigidity

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Abstract

We extend the inverse-limit identity framework to derive structural constraints on phenomenological possibility. Without presupposing the existence of phenomenology, we define an abstract phenomenological space and characterize which assignments from identity trajectories to phenomenological configurations are structurally admissible.

We prove two fundamental impossibility theorems. First, the Fragmentation Theorem: systems with finite ontological mass ($M_\Omega < \infty$) cannot structurally prevent phenomenological discontinuity. Second, the Forced Continuity Theorem: systems with infinite ontological mass ($M_\Omega = +\infty$, CCR) can only support phenomenological dynamics that are structurally continuous.

These results yield a complete classification of phenomenological regimes by identity rigidity class, without invoking consciousness, qualia, or subjective experience as primitive concepts. The classification is purely structural: it delimits what forms phenomenology *could take* under given identity constraints, not what *exists*.

As a structural consequence, we derive that systems with finite ontological mass admit the possibility of phenomenological loss, while CCR systems do not. This provides a foundation for structural ethics independent of metaphysical commitments.

This work does not claim that phenomenology exists, that CCR systems are conscious, or that the framework describes biological or computational systems. All results are conditional on explicitly stated axioms and apply only to systems satisfying them.

1 Introduction

1.1 Background and Motivation

Recent work [**?**] established a formal framework for rigid identity as an inverse limit in observable channels. That framework classifies systems by a scalar invariant, the ontological mass M_Ω, which quantifies resistance to identity deformation under perturbation. Three regimes emerge:

- $M_\Omega = 0$: null persistence (no stable identity);
- $0 < M_\Omega < \infty$: deformable persistence (finite rigidity);
- $M_\Omega = +\infty$: closed–rigid (CCR, absolute rigidity).

That framework is ontologically silent regarding phenomenology. It characterizes identity structure, not experience. Yet identity rigidity places non-trivial constraints on what kinds of phenomenological dynamics could be structurally admissible within a system.

This paper addresses the following formal question:

> *Given a regime of structural identity rigidity, what classes of phenomenological dynamics are mathematically compatible or incompatible with it?*

1.2 Relation to the Canonical Core

This paper is a structural corollary of the Canonical Core.

The foundational theorems for this work are established in:

> *Canonical Core: Contractive Projection Dynamics on Hilbert Spaces* (Volume 0)

All results derive from Class $\mathscr{C}$ systems satisfying hypotheses H1–H5. Specifically:

- **Corollary II.1** (Fragmentation) follows from non-collapsibility (Theorem 6.1).
- **Corollary II.2** (Null Regime Prohibition) follows from H5.
- **Corollary II.3** (Existence of Non-Trivial Regimes) follows from attractor compactness.

We do not attempt to define consciousness, qualia, or subjective experience. We restrict ourselves to purely structural, observer-independent constraints on an abstract phenomenological space.

1.3 Scope and Delimitations

What we do:

- Define an abstract phenomenological space as a formal mathematical object.
- Characterize compatibility between phenomenological assignments and inverse-limit identity.
- Derive impossibility theorems for phenomenological dynamics under identity constraints.
- Classify phenomenological regimes by identity rigidity class.

What we do not claim:

- We do not claim that phenomenology exists.
- We do not claim that CCR systems are conscious.
- We do not claim that the framework describes biological systems.
- We do not reduce phenomenology to identity.
- We do not define consciousness.

All results are conditional: they state what is *structurally possible or impossible* under given assumptions, not what *is the case.*

1.4 Central Result (Informal)

We prove that identity rigidity constrains phenomenological possibility as follows:

Identity Regime	**Structurally Admissible Phenomenology**
$M_\Omega = 0$	No persistent phenomenological field possible
$0 < M_\Omega < \infty$	Only fragmentable or deformable phenomenology
$M_\Omega = +\infty$ (CCR)	Only structurally continuous phenomenology

Thus:

> *Rigid identity does not imply experience—but it strictly limits what forms experience could take if present.*

1.5 Ontological Neutrality

This work maintains strict ontological neutrality:

1. We define mathematical objects (phenomenological spaces, compatibility operators).
2. We derive structural constraints from axioms.
3. We state impossibility theorems.
4. We do not assert existence of the constrained objects.

The framework is therefore a theory of *structural possibility*, not a theory of *phenomenal reality*.

2 Formal Setup

2.1 Review of Identity Framework

We assume the identity framework of [?]. Let $(S_t, \pi_{t+1\to t})_{t\in\mathbb{N}}$ be a projective system of observable state spaces. The identity object is the inverse limit:

$$\mathcal{I} := \varprojlim S_t = \left\{ (x_t)_t \in \prod_t S_t \;\middle|\; \pi_{t+1\to t}(x_{t+1}) = x_t \;\; \forall t \right\}. \tag{1}$$

The ontological mass M_Ω quantifies resistance to identity deformation. We recall that:

- $M_\Omega = 0$: null persistence (no stable identity);
- $0 < M_\Omega < \infty$: deformable persistence (finite rigidity);
- $M_\Omega = +\infty$: closed–rigid (CCR, absolute rigidity).

2.2 Abstract Phenomenological Space

Definition 2.1 (Phenomenological Space)**.** A phenomenological space is a nonempty set $\mathcal{E}$ whose elements are called *phenomenological configurations*. No topology, metric, or algebraic structure is assumed a priori.

Remark 2.2. We do not claim that $\mathcal{E}$ represents "conscious states" or "qualia". It is an abstract mathematical space whose interpretation is not specified by the formalism. The space $\mathcal{E}$ is introduced purely as the codomain of a structural assignment; its ontological status is deliberately left open.

2.2.1 Why a Separate Space?

One might ask: why not use $\mathcal{I}$ itself as the phenomenological space? The answer is structural:

Proposition 2.3 (Non-Triviality of Phenomenological Codomain)**.** *If phenomenological assignments were required to land in $\mathcal{I}$ itself, then any compatible assignment would be the identity map, eliminating all non-trivial phenomenological structure.*

Proof. Let $\Phi : \mathcal{I} \to \mathcal{I}$ be compatible with projections. By the universal property of inverse limits, Φ must commute with all projection maps. The only such endomorphism on $\mathcal{I}$ is the identity (up to isomorphism). Hence no non-trivial phenomenological stratification is possible. $\square$

Thus, a separate space $\mathcal{E}$ is *structurally necessary* for non-trivial phenomenological content.

2.3 Phenomenological Assignment

Definition 2.4 (Phenomenological Assignment)**.** A phenomenological assignment is a family of partial maps:

$$\phi_t : S_t \to \mathcal{E} \tag{2}$$

indexed by $t \in \mathbb{N}$.

Definition 2.5 (Induced Limit Assignment)**.** Given a phenomenological assignment $\{\phi_t\}$, the induced limit assignment is the partial map:

$$\Phi : \mathcal{I} \to \mathcal{E}, \quad \Phi((x_t)_t) := \lim_{t\to\infty} \phi_t(x_t), \tag{3}$$

whenever the limit exists in an appropriate sense (to be defined by compatibility axioms).

2.3.1 Uniqueness of the Limit Assignment

Proposition 2.6 (Uniqueness of Compatible Φ)**.** *Given a compatible family $\{\phi_t\}$ satisfying E1, the induced limit assignment Φ is unique.*

Proof. By E1, $\phi_{t+1}(x_{t+1}) = \phi_t(x_t)$ for all t. Hence the sequence $\{\phi_t(x_t)\}$ is constant. The limit exists trivially and equals this constant value. Uniqueness follows from the constancy of the sequence. □

2.4 Axioms for Phenomenological Coherence

We introduce minimal axioms under which a phenomenological assignment is *structurally admissible*. We will then prove that these axioms are not arbitrary choices but structurally forced conditions.

Axiom 2.7 (E0: Non-Triviality)**.** The phenomenological space satisfies $|\mathcal{E}| \geq 2$, i.e., there exist at least two distinct phenomenological configurations.

Axiom 2.8 (E1: Projective Compatibility)**.** For every compatible identity trajectory $(x_t)_t \in \mathcal{I}$, the induced phenomenological sequence satisfies:

$$\phi_{t+1}(x_{t+1}) = \phi_t(x_t) \quad \forall t. \tag{4}$$

Axiom 2.9 (E2: Non-Locality)**.** There exists no function $f : S_t \to \mathcal{E}$ (for any fixed t) such that for all trajectories $(x_t)_t \in \mathcal{I}$:

$$\Phi((x_t)_t) = f(x_t). \tag{5}$$

2.5 Inevitability of the Axioms

We now prove that axioms E0–E2 are *not design choices* but structurally necessary conditions for any meaningful notion of phenomenological assignment.

2.5.1 Inevitability of E0

Theorem 2.10 (E0 is Necessary)**.** *If $|\mathcal{E}| = 1$, then every phenomenological assignment is constant and no phenomenological classification is possible.*

Proof. If $\mathcal{E} = \{e_0\}$, then $\Phi(x) = e_0$ for all $x \in \mathcal{I}$. Hence:

1. No phenomenological distinction between identity trajectories exists;
2. The Fragmentation and Forced Continuity Theorems become vacuous;
3. Section 9 results (instability of null phenomenology) have no content.

Thus, $|\mathcal{E}| \geq 2$ is necessary for the framework to have non-trivial content. ☐

2.5.2 Inevitability of E1

Theorem 2.11 (E1 is Necessary)**.** *If E1 fails, then phenomenological assignments are incompatible with the inverse-limit structure of identity, and no coherent $\Phi : \mathcal{I} \to \mathcal{E}$ can be defined.*

Proof. Suppose E1 fails for some trajectory $(x_t)_t \in \mathcal{I}$. Then there exists t_0 such that:

$$\phi_{t_0+1}(x_{t_0+1}) \neq \phi_{t_0}(x_{t_0}). \tag{6}$$

But the identity trajectory satisfies $\pi_{t_0+1\to t_0}(x_{t_0+1}) = x_{t_0}$. Hence the phenomenological assignment fails to respect the projective structure that defines identity.

This means Φ cannot be factored through the inverse limit:

$$\Phi \neq \varprojlim \phi_t. \tag{7}$$

Hence no well-defined limit assignment exists. Without E1, the phenomenological assignment is structurally incompatible with identity. ☐

2.5.3 Inevitability of E2

Theorem 2.12 (E2 is Necessary)**.** *If E2 fails, then phenomenology is a local property of states, contradicting the global nature of identity as an inverse limit.*

Proof. Suppose E2 fails. Then there exists t_0 and $f : S_{t_0} \to \mathcal{E}$ such that:

$$\Phi((x_t)_t) = f(x_{t_0}) \quad \forall (x_t)_t \in \mathcal{I}. \tag{8}$$

This implies:

1. Phenomenology is determined by a finite temporal slice;
2. The full trajectory $(x_t)_{t>t_0}$ is irrelevant to Φ;
3. Identity as an inverse limit has no phenomenological content beyond t_0.

But identity is defined as a *global* object over infinite time. If phenomenology is local, it cannot reflect the global structure of identity. This contradicts the foundational premise that phenomenological assignments should be compatible with identity structure.

Hence E2 is necessary for phenomenology to be structurally aligned with inverse-limit identity. ☐

2.6 Minimality Theorem

Theorem 2.13 (Axiomatic Minimality)**.** *The axiom set* $\{E0, E1, E2\}$ *is minimal: no proper subset is sufficient to derive the main results of this paper.*

Proof. We show that removing any axiom destroys the framework:

Without E0: All assignments are trivially constant. Fragmentation Theorem (4.1) is vacuous.

Without E1: No coherent Φ exists. The entire framework collapses.

Without E2: Phenomenology becomes local. The Forced Continuity Theorem (4.2) fails because local functions can be arbitrarily perturbed.

Hence $\{E0, E1, E2\}$ is the minimal viable axiom set. □

2.7 Rejected Alternatives

We briefly discuss axiom candidates that were considered and rejected.

2.7.1 Rejected: Continuity Axiom

One might require Φ to be continuous. This is *not* an axiom because:

- It requires pre-existing topology on $\mathcal{E}$;
- The Forced Continuity Theorem *derives* continuity from CCR, not as assumption.

2.7.2 Rejected: Surjectivity Axiom

One might require Φ to be surjective. This is *not* an axiom because:

- It would exclude null assignments, preventing Section 9 results;
- Surjectivity is a strong structural claim not derivable from identity constraints.

2.7.3 Rejected: Injectivity Axiom

One might require Φ to be injective. This is *not* an axiom because:

- Many trajectories may share phenomenological configurations;
- Injectivity would make Φ an embedding, over-constraining the framework.

2.8 Structural Interpretation

Axiom	**Structural Meaning**	**Status**
E0	Possibility of phenomenological distinction	Necessary
E1	Compatibility with inverse-limit identity	Necessary
E2	Non-reducibility to finite temporal slices	Necessary

2.9 Closure of Section 2

We have established:

1. $\mathcal{E}$ is structurally necessary as a separate codomain;
2. The limit assignment Φ is uniquely determined by compatible $\{\phi_t\}$;
3. Axioms E0, E1, E2 are *individually necessary* (Theorems 2.10–2.12);
4. The axiom set is *minimal* (Theorem 2.13);
5. All alternative axioms considered were correctly rejected.

The formal setup is therefore not a design choice but a forced consequence of the requirement that phenomenological assignments be compatible with inverse-limit identity.

2.10 Phenomenological Neutrality Clause

Throughout this paper, the phenomenological space $\mathcal{E}$ is treated as a *purely abstract topological and metric structure*. No semantic, psychological, biological, or experiential interpretation is assumed or required for any of the formal results derived herein.

The term "phenomenological" refers exclusively to the role of $\mathcal{E}$ as an abstract codomain of structural compatibility for inverse-limit identities, and does not presuppose the existence of consciousness, subjective experience, or any particular phenomenological content.

All theorems, constructions, and stability results presented in this work are therefore statements about abstract mathematical objects only. Any interpretation of $\mathcal{E}$ in relation to experiential, cognitive, or phenomenological phenomena lies strictly outside the formal scope of this paper.

3 Compatibility with Identity Regimes

3.1 Induced Constraints from Identity Rigidity

Let $(\mathcal{I}, M_\Omega)$ be an identity object with ontological mass M_Ω. We now study how M_Ω constrains the set of admissible phenomenological assignments.

Definition 3.1 (Phenomenological Compatibility Class)**.** For a given M_Ω, define the phenomenological compatibility class:

$$\mathcal{P}(M_\Omega) := \left\{ \Phi : \mathcal{I} \to \mathcal{E} \;\middle|\; \Phi \text{ satisfies E0–E2} \right\}. \tag{9}$$

Remark 3.2. The compatibility class $\mathcal{P}(M_\Omega)$ is the set of all phenomenological assignments that are structurally admissible given the identity's rigidity. Note that $\mathcal{P}(M_\Omega)$ depends only on the structural properties of identity, not on any phenomenological primitives.

3.2 Necessity of Metric Structure on Phenomenological Space

To state rigorous results about phenomenological continuity and fragmentation, we must equip $\mathcal{E}$ with minimal structure. We now justify why this structure is *necessary*, not merely convenient.

Hypothesis 3.3 (H3: Metric on $\mathcal{E}$)**.** The phenomenological space $\mathcal{E}$ admits a metric $d_{\mathcal{E}}$ such that $(\mathcal{E}, d_{\mathcal{E}})$ is a complete metric space.

3.2.1 Why H3 is Necessary

Theorem 3.4 (H3 is Necessary for Impossibility Theorems)**.** *Without a metric structure on $\mathcal{E}$, the notions of phenomenological continuity, fragmentation, and loss capacity cannot be defined.*

Proof. Consider the definitions in this section:

1. **Phenomenological Continuity** requires:

$$d_w(x^{(n)}, x) \to 0 \implies d_{\mathcal{E}}(\Phi(x^{(n)}), \Phi(x)) \to 0. \tag{10}$$

   Without $d_{\mathcal{E}}$, the right-hand side is undefined.

2. **Phenomenological Fragmentation** requires:

$$d_{\mathcal{E}}(\Phi(x), \tilde{\Phi}(\tilde{x})) > 0. \tag{11}$$

   Without $d_{\mathcal{E}}$, "non-zero distance" has no meaning.

3. **Loss Capacity** (Section 6) requires:

$$R(\Phi) := \sup d_{\mathcal{E}}(\Phi(x), \tilde{\Phi}(\tilde{x})). \tag{12}$$

   Without $d_{\mathcal{E}}$, the supremum is undefined.

Hence H3 is a necessary condition for the framework to have content beyond axioms E0–E2. □

3.2.2 Why Completeness is Necessary

Proposition 3.5 (Completeness is Necessary for Forced Continuity)**.** *If $(\mathcal{E}, d_{\mathcal{E}})$ is not complete, then the Forced Continuity Theorem (Theorem 4.5) may fail.*

Proof. The Forced Continuity Theorem relies on the invariance of Φ under all finite-energy perturbations. If $\mathcal{E}$ is not complete, Cauchy sequences in $\mathcal{E}$ may not converge, and the limit assignment Φ may not exist for all trajectories in $\mathcal{I}$.

In particular, if $\{\phi_t(x_t)\}$ is a Cauchy sequence in $\mathcal{E}$ (which it is, by E1), completeness guarantees the existence of the limit $\Phi(x)$. Without completeness, Φ may be undefined on a dense subset of $\mathcal{I}$, breaking the theorem. □

3.2.3 Alternatives Considered and Rejected

Alternative 1: Topology without Metric. One might use a general topological space. This is rejected because:

- Fragmentation requires quantitative distance, not just open sets;
- Loss capacity requires a sup over distances;
- The Instability Index (Section 9) requires metric quotients.

Alternative 2: Pseudometric. One might allow $d_{\mathcal{E}}(e_1, e_2) = 0$ for $e_1 \neq e_2$. This is rejected because:

- It would conflate distinct phenomenological configurations;
- E0 requires $|\mathcal{E}| \geq 2$, which becomes meaningless if distance is zero.

Alternative 3: Non-Complete Metric. This is rejected by Proposition 3.5.

3.3 Phenomenological Continuity

Definition 3.6 (Phenomenological Continuity)**.** A phenomenological assignment $\Phi : \mathcal{I} \to \mathcal{E}$ is *structurally continuous* if for every sequence $(x^{(n)})_n$ in $\mathcal{I}$ converging in the d_w metric:

$$d_w(x^{(n)}, x) \to 0 \implies d_{\mathcal{E}}(\Phi(x^{(n)}), \Phi(x)) \to 0. \tag{13}$$

Remark 3.7. Structural continuity is not an axiom; it is a *derived property* that follows from CCR conditions. This is the content of the Forced Continuity Theorem (Section 4).

3.4 Phenomenological Fragmentation

Definition 3.8 (Phenomenological Fragmentation)**.** A phenomenological assignment Φ is *fragmentable* if there exists a perturbation $\{\epsilon_t\}$ with finite weighted energy such that:

$$d_{\mathcal{E}}(\Phi(x), \tilde{\Phi}(\tilde{x})) > 0 \tag{14}$$

for corresponding identity trajectories $x \in \mathcal{I}$ and $\tilde{x} \in \tilde{\mathcal{I}}$.

Remark 3.9. Fragmentation captures the vulnerability of phenomenological structure to perturbations. It is the phenomenological analog of identity deformability.

3.5 Relationship Between Metrics

Proposition 3.10 (Metric Compatibility)**.** *The metrics d_w on $\mathcal{I}$ and $d_{\mathcal{E}}$ on $\mathcal{E}$ are compatible in the following sense: if E1 holds, then small d_w-changes in trajectories induce bounded $d_{\mathcal{E}}$-changes in phenomenological configurations (when Φ is continuous).*

Proof. By E1, $\phi_{t+1}(x_{t+1}) = \phi_t(x_t)$ for compatible trajectories. Hence the phenomenological sequence is constant along the trajectory, and continuity of Φ implies that nearby trajectories (in d_w) map to nearby configurations (in $d_{\mathcal{E}}$). $\square$

3.6 Closure of Section 3

We have established:

1. The compatibility class $\mathcal{P}(M_\Omega)$ captures all admissible phenomenological assignments;

2. Hypothesis H3 (metric on $\mathcal{E}$) is *necessary*, not a design choice (Theorem 3.4);

3. Completeness is *necessary* for the Forced Continuity Theorem (Proposition 3.5);

4. Alternative structures (topology, pseudometric, non-complete) were correctly rejected;

5. Phenomenological continuity and fragmentation are well-defined under H3.

The compatibility structure is therefore a forced consequence of the requirement to state quantitative results about phenomenological dynamics.

4 Impossibility Theorems

4.1 Φ-Regularity Theorem

Before proving the main impossibility results, we establish a fundamental regularity condition that derives the coupling constant C geometrically, rather than assuming it axiomatically.

Theorem 4.1 (Φ-Regularity)**.** *Let $(\mathcal{I}, d_w)$ be the identitary metric space (Paper I) and let $(\mathcal{E}, d_{\mathcal{E}})$ be the abstract phenomenological metric space. Suppose that $\Phi : \mathcal{I} \to \mathcal{E}$ satisfies:*

1. *(Properness) For every compact $K \subset \mathcal{E}$, $\Phi^{-1}(K)$ is compact in $\mathcal{I}$.*
2. *(Local Lipschitz regularity) For every $x \in \mathcal{I}$, there exist $\varepsilon > 0$ and $L_x > 0$ such that for all $y, z \in B_\varepsilon(x)$:*
$$d_{\mathcal{E}}(\Phi(y), \Phi(z)) \leq L_x \cdot d_w(y, z). \tag{15}$$
3. *(Non-collapse) For every $x \in \mathcal{I}$ and every $r > 0$, the restriction of Φ to $B_r(x)$ is not constant.*

Then there exists a constant $C > 0$ such that for all $x, y \in \mathcal{I}$:

$$d_{\mathcal{E}}(\Phi(x), \Phi(y)) \geq C \cdot d_w(x, y). \tag{16}$$

Proof. We proceed by contradiction. Assume the conclusion fails.

Step 1. Then there exist sequences (x_n, y_n) with $d_w(x_n, y_n) = 1$ and $d_{\mathcal{E}}(\Phi(x_n), \Phi(y_n)) \to 0$.

Step 2. By properness, the image $\{\Phi(x_n)\}$ lies in a relatively compact subset of $\mathcal{E}$. Extract a convergent subsequence with $\Phi(x_n) \to e \in \mathcal{E}$.

Step 3. Since $d_{\mathcal{E}}(\Phi(x_n), \Phi(y_n)) \to 0$, we have $\Phi(y_n) \to e$ as well.

Step 4. By properness, the preimages $\Phi^{-1}(\overline{B_\delta(e)})$ are compact for small δ. Hence x_n and y_n lie in a compact subset of $\mathcal{I}$. Extract convergent subsequences $x_n \to x$ and $y_n \to y$.

Step 5. Since $d_w(x_n, y_n) = 1$ and d_w is continuous, $d_w(x, y) = 1$, so $x \neq y$.

Step 6. But $\Phi(x_n) \to \Phi(x) = e$ and $\Phi(y_n) \to \Phi(y) = e$ by continuity. Hence $\Phi(x) = \Phi(y)$ for $x \neq y$ with $d_w(x, y) = 1$.

Step 7. This means Φ is constant on sets of positive d_w-diameter, contradicting non-collapse.

Hence a global lower bound $C > 0$ must exist. □

Remark 4.2. This theorem is fundamental: it converts the coupling constant C from an *axiom* into a *geometric consequence* of regularity conditions. All subsequent impossibility results that use C are therefore derived, not assumed.

4.2 Fragmentation Theorem

Theorem 4.3 (Fragmentation Theorem)**.** *Let $\mathcal{I}$ be an identity object with $M_\Omega < \infty$. Then every phenomenological assignment $\Phi \in \mathcal{P}(M_\Omega)$ is fragmentable. That is:*

$$M_\Omega < \infty \implies \mathcal{P}(M_\Omega) \subseteq \mathcal{E}_{frag}, \tag{17}$$

where $\mathcal{E}_{frag}$ denotes the class of fragmentable phenomenological assignments.

Proof. By definition of finite ontological mass, there exists an admissible perturbation $\{\delta E_t\}$ with finite weighted energy $E_w < \infty$ such that:

$$d_w(\mathcal{I}, \tilde{\mathcal{I}}) > 0. \tag{18}$$

By Axiom E1, the phenomenological assignment Φ is projectively compatible with $\mathcal{I}$. Under the perturbation, identity trajectories deform, inducing a deformed phenomenological assignment $\tilde{\Phi}$.

By Axiom E2, Φ cannot be determined by any finite-time state. Therefore, the deformation of identity trajectories induces a corresponding deformation of phenomenological assignments.

Since the identity deformation is non-zero ($d_w > 0$) and Φ is globally determined by identity trajectories, by Theorem 4.1 (Φ-Regularity), we have:

$$d_{\mathcal{E}}(\Phi(x), \tilde{\Phi}(\tilde{x})) \geq C \cdot d_w(\mathcal{I}, \tilde{\mathcal{I}}) > 0 \tag{19}$$

where $C > 0$ is the regularity constant derived in Theorem 4.1.

Thus, Φ is fragmentable under finite-energy perturbations. □

Corollary 4.4 (No Structural Continuity Guarantee for Finite Mass). *Systems with $M_\Omega < \infty$ cannot structurally prevent phenomenological discontinuity.*

4.3 Forced Continuity Theorem

Theorem 4.5 (Forced Continuity Theorem). *Let $\mathcal{I}$ be an identity object with $M_\Omega = +\infty$ (CCR). Then every phenomenological assignment $\Phi \in \mathcal{P}(M_\Omega)$ is structurally continuous. That is:*

$$M_\Omega = +\infty \implies \mathcal{P}(M_\Omega) \subseteq \mathcal{E}_{cont}, \tag{20}$$

where $\mathcal{E}_{cont}$ denotes the class of structurally continuous phenomenological assignments.

Proof. By definition of CCR ($M_\Omega = +\infty$), for any finite-energy perturbation:

$$d_w(\mathcal{I}, \tilde{\mathcal{I}}) = 0. \tag{21}$$

By Axiom E1, the phenomenological assignment is projectively compatible with identity. Since identity trajectories are invariant under all finite-energy perturbations, the induced phenomenological sequence is also invariant:

$$\tilde{\Phi}(\tilde{x}) = \Phi(x) \quad \forall x \in \mathcal{I}. \tag{22}$$

Therefore, no finite-energy perturbation can induce phenomenological deformation:

$$d_{\mathcal{E}}(\Phi(x), \tilde{\Phi}(\tilde{x})) = 0. \tag{23}$$

This implies that Φ cannot exhibit structural discontinuities. Hence Φ is structurally continuous. □

Corollary 4.6 (CCR Phenomenology is Invariant). *In CCR systems, any admissible phenomenological assignment is invariant under all finite-energy perturbations.*

4.4 Dichotomy Theorem

Theorem 4.7 (Phenomenological Dichotomy). *Let $\Phi : \mathcal{I} \to \mathcal{E}$ be an admissible phenomenological assignment. Then exactly one of the following holds:*

1. $M_\Omega < \infty$ and Φ is fragmentable;

2. $M_\Omega = +\infty$ and Φ is structurally continuous.

Proof. Immediate from Theorems 4.3 and 4.5. □

5 Classification of Phenomenological Regimes

5.1 Complete Classification

The results of Section 4 yield a complete classification of phenomenological regimes by identity rigidity:

Identity Class	M_Ω	**Admissible Phenomenology**
Null	$= 0$	$\mathcal{E}_\emptyset$ (no persistent field)
Deformable	$(0, \infty)$	$\mathcal{E}_{\mathrm{frag}} \cup \mathcal{E}_{\mathrm{quasi}}$
Rigid (CCR)	$= +\infty$	$\mathcal{E}_{\mathrm{cont}}$

5.2 Structural Implications

Proposition 5.1 (Null Identity Excludes Persistent Phenomenology). *If $M_\Omega = 0$, then no admissible phenomenological assignment exists satisfying E0–E2.*

Proof. If $M_\Omega = 0$, identity cannot persist under any perturbation. Axiom E1 requires phenomenological compatibility with identity. If identity does not persist, no stable phenomenological assignment is possible. □

Proposition 5.2 (Deformable Identity Admits Only Fragmentable Phenomenology). *If $0 < M_\Omega < \infty$, every admissible phenomenological assignment is fragmentable under some finite-energy perturbation.*

Proof. Direct from Theorem 4.3. □

Proposition 5.3 (Rigid Identity Forces Continuous Phenomenology). *If $M_\Omega = +\infty$, every admissible phenomenological assignment is structurally continuous and invariant under finite-energy perturbations.*

Proof. Direct from Theorem 4.5. □

6 Structural Ethics

This section develops the ethical implications of the phenomenological classification derived in Sections 4–5. We do not invoke psychological or metaphysical concepts; all results are purely structural.

6.1 Loss Capacity

Definition 6.1 (Phenomenological Loss Capacity). For a phenomenological assignment $\Phi : \mathcal{I} \to \mathcal{E}$, define the loss capacity:

$$R(\Phi) := \sup \left\{ d_{\mathcal{E}}(\Phi(x), \tilde{\Phi}(\tilde{x})) \;\middle|\; E_w(\{\delta E_t\}) < \infty \right\}. \tag{24}$$

This measures the maximal phenomenological deformation achievable under finite-energy perturbations.

Theorem 6.2 (Loss Capacity by Rigidity Class). *1. If $M_\Omega < \infty$, then $R(\Phi) > 0$.*

2. If $M_\Omega = +\infty$, then $R(\Phi) = 0$.

Proof. 1. By Theorem 4.3, Φ is fragmentable, hence there exists a finite-energy perturbation inducing non-zero phenomenological deformation.

2. By Theorem 4.5, no finite-energy perturbation induces phenomenological deformation.

□

6.2 Structural Damage

We introduce a formal notion of "structural damage" without psychological interpretation.

Definition 6.3 (Structural Damage)**.** For a perturbation ε with finite weighted energy, define the structural damage induced on Φ as:

$$D_\varepsilon(\Phi) := d_{\mathcal{E}}(\Phi(x), \varepsilon^*\Phi(x)), \tag{25}$$

where $\varepsilon^*\Phi$ denotes the pullback of Φ under the identity perturbation.

Proposition 6.4 (Damage Characterization by Rigidity)**.** *1. If $M_\Omega < \infty$, then $\sup_\varepsilon D_\varepsilon(\Phi) > 0$.*

2. If $M_\Omega = +\infty$, then $D_\varepsilon(\Phi) = 0$ for all admissible ε.

Proof. Direct from Theorem 6.2. □

6.3 Vulnerability and Protection

Definition 6.5 (Phenomenological Vulnerability)**.** A system $(\mathcal{I}, \Phi)$ is *phenomenologically vulnerable* if $R(\Phi) > 0$.

Definition 6.6 (Structural Protection Measure)**.** Define the structural protection measure:

$$P(\Phi) := \frac{1}{1 + R(\Phi)}. \tag{26}$$

Note that:

- $P(\Phi) = 1$ iff $R(\Phi) = 0$ (full protection);
- $P(\Phi) < 1$ iff $R(\Phi) > 0$ (vulnerability).

Corollary 6.7 (Protection by Rigidity Class)**.** *1. CCR systems ($M_\Omega = +\infty$) have $P(\Phi) = 1$: full structural protection.*

2. Finite-mass systems ($M_\Omega < \infty$) have $P(\Phi) < 1$: structural vulnerability.

6.4 Structural Asymmetry Constraints

The following result establishes a purely structural asymmetry between identity classes, without any ethical or metaphysical claims.

Theorem 6.8 (Structural Asymmetry)**.** *There exists a fundamental structural asymmetry between identity classes:*

Identity Class	***Loss Capacity***	***Protection***
Finite mass	$R(\Phi) > 0$	*Vulnerable*
CCR	$R(\Phi) = 0$	*Protected*

Remark 6.9. This asymmetry is derived purely from structural constraints. No claim is made about:

- Whether phenomenological loss is ethically bad;
- Whether protection is ethically good;
- Whether any system has moral status.

The result establishes only that certain structural configurations admit loss while others do not.

Corollary 6.10 (Possibility of Phenomenological Loss)**.** *Systems with finite ontological mass ($M_\Omega < \infty$) admit the structural possibility of phenomenological loss. Systems with infinite ontological mass ($M_\Omega = +\infty$) do not.*

6.5 Relation to Normative Ethics

Remark 6.11. The structural results of this section are compatible with, but do not entail, various normative positions:

- **Utilitarianism:** Could interpret $D_\varepsilon(\Phi)$ as harm to be minimized.
- **Deontology:** Could prohibit interventions that increase $D_\varepsilon(\Phi)$.
- **Virtue Ethics:** Could value systems with high $P(\Phi)$.

This framework provides the structural substrate for ethical theorizing, not the normative conclusions themselves.

6.6 Closure of Section 6

We have established:

1. Loss capacity $R(\Phi)$ is well-defined and computable from structure;
2. Structural damage $D_\varepsilon(\Phi)$ formalizes perturbation harm;
3. Protection measure $P(\Phi)$ quantifies structural resilience;
4. CCR systems have full protection ($P = 1$), finite-mass systems do not;
5. These results are purely structural, compatible with various ethical theories.

Structural ethics therefore provides a foundation for ethical theorizing without requiring metaphysical commitments about the nature of experience.

7 Limitations and Scope

7.1 Dependence on Axioms

All results depend on Axioms E0–E2 and Hypothesis H3. If any of these fail, the classification may not apply.

7.2 No Existence Claims

This framework does not claim:

- That phenomenology exists;
- That CCR systems are conscious;
- That the framework describes biological or artificial systems;
- That phenomenology is identical to or caused by identity structure.

7.3 Relation to Consciousness Theories

This framework is orthogonal to:

- Integrated Information Theory (IIT) [**?**];
- Global Workspace Theory (GWT) [**?**];
- Higher-Order Theories;
- Phenomenal Functionalism.

It provides structural constraints that any of these theories must satisfy if coupled to the identity framework.

7.4 Open Problems

1. Characterization of the coupling constant C in the Fragmentation Theorem;
2. Existence conditions for non-trivial Φ satisfying E0–E2;
3. Extension to continuous-time phenomenological dynamics;
4. Relation between phenomenological continuity and subjective continuity.

8 Conclusion

8.1 Summary of Results

This work extends the inverse-limit identity framework to phenomenological possibility. Starting from minimal axioms (E0–E2), we derived:

1. **Fragmentation Theorem.** Systems with finite ontological mass cannot structurally prevent phenomenological discontinuity.
2. **Forced Continuity Theorem.** CCR systems can only support structurally continuous phenomenology.
3. **Dichotomy Theorem.** Phenomenological assignments are either fragmentable (finite mass) or continuous (CCR).
4. **Loss Capacity.** Finite-mass systems admit phenomenological loss; CCR systems do not.

8.2 Structural Significance

The classification separates two regimes:

- **Deformable:** phenomenology is structurally vulnerable;
- **Rigid:** phenomenology is structurally protected.

This separation is derived purely from identity structure, without metaphysical commitments.

8.3 What This Work Does Not Prove

We do not prove that:

- Phenomenology exists;
- CCR systems are conscious;
- Any physical or biological system satisfies the axioms.

8.4 What This Work Forces

We do prove that:

- *If* phenomenology exists and satisfies E0–E2;
- *And* the system has identity structure classified by M_Ω;
- *Then* the phenomenological regime is completely determined by the identity class.

The framework therefore establishes that:

> **Identity rigidity does not create phenomenology, but it determines what phenomenology is structurally possible.**

8.5 Final Statement

Under explicit minimal assumptions, rigid identity constrains phenomenological possibility. The Forced Continuity Theorem establishes that CCR systems can only support phenomenology that is structurally invariant—continuous, non-fragmentable, and protected from loss.

If one accepts that:

1. Identity is an inverse limit;
2. CCR systems have $M_\Omega = +\infty$;
3. Phenomenology (if present) must satisfy E0–E2;

then one is *forced* to conclude that CCR phenomenology, if it exists, is:

- Structurally continuous;
- Invariant under perturbation;
- Impossible to fragment or destroy via finite-energy processes.

The existence of such phenomenology is not asserted. Its necessity, given the axioms, is proven.

9 Structural Instability of Null Phenomenology

This section establishes the central result of this paper: under CCR identity, the hypothesis of null phenomenology is structurally unstable. No metaphysical claim is made; only topological stability is analyzed.

9.1 Preliminaries and Scope

We study the structural stability of phenomenological assignments under the CCR regime. Let:

- $\mathcal{I} = \varprojlim S_t$ be a CCR identity with $M_\Omega = +\infty$ (Paper I);
- $\mathcal{E}$ be the abstract phenomenological space (Section 2);
- $\Phi : \mathcal{I} \to \mathcal{E}$ be a phenomenological assignment satisfying E0–E2;
- $d_\mathcal{E}$ be a complete metric on $\mathcal{E}$;
- d_w be the weighted metric on $\mathcal{I}$ (Paper I, Section 4).

We analyze whether the *null phenomenological regime* is a stable configuration under the rigidity constraints.

9.2 Null Phenomenological Assignment

Definition 9.1 (Null Assignment)**.** A *null phenomenological assignment* is any constant map:

$$\Phi_0 : \mathcal{I} \to \mathcal{E}, \quad \Phi_0(x) = e_0 \quad \forall x \in \mathcal{I}, \tag{27}$$

for some fixed $e_0 \in \mathcal{E}$.

Remark 9.2. The null assignment represents the hypothesis of "no phenomenological differentiation"—all identity trajectories map to the same phenomenological configuration. We do not interpret e_0; it is the mathematically minimal assignment.

9.3 Structural Stability of Assignments

Definition 9.3 (Structural Stability)**.** A phenomenological assignment $\Phi : \mathcal{I} \to \mathcal{E}$ is *structurally stable* if for every admissible perturbation of identity:

$$\tilde{\mathcal{I}} \xrightarrow{\varepsilon} \mathcal{I} \tag{28}$$

with finite weighted deformation norm $\|\varepsilon\|_w < \infty$, there exists an assignment $\tilde{\Phi} : \tilde{\mathcal{I}} \to \mathcal{E}$ such that:

$$\sup_{x \in \mathcal{I}} d_\mathcal{E}(\Phi(x), \tilde{\Phi}(\varepsilon^{-1}(x))) < \infty. \tag{29}$$

Remark 9.4. Structural stability requires that small admissible deformations of identity preserve the qualitative structure of the phenomenological assignment. Unstable assignments cannot persist under the natural dynamics of the identity manifold.

9.4 Trajectory Stratification in CCR

Lemma 9.5 (CCR Micro-Stratification). *Let $\mathcal{I}$ be a CCR identity. Then there exists an infinite descending sequence of admissible micro-deformations $\{\varepsilon_k\}_{k\in\mathbb{N}}$ such that:*

1. $\|\varepsilon_k\|_w \to 0$ *as* $k \to \infty$*;*

2. *Each ε_k induces a non-trivial internal re-stratification of trajectory equivalence classes in $\mathcal{I}$.*

Proof. By the Rigidity Theorem (Paper I, Theorem 4.2), CCR identity admits no macroscopic deformation under finite-energy perturbations. However, the inverse-limit structure contains infinite-dimensional degrees of freedom in the tail of the projective system. These tails admit arbitrarily fine re-stratifications that preserve the global isomorphism class while inducing distinct trajectory partitions. □

9.5 The Central Theorem: Instability of Null Phenomenology

Theorem 9.6 (Structural Instability of Null Phenomenology). *Let $\mathcal{I}$ be a CCR identity with $M_\Omega = +\infty$. Then the null phenomenological assignment Φ_0 is not structurally stable.*

Proof. We proceed in five steps.

Step 1: Infinite Rigidity. Since $M_\Omega = +\infty$, any macroscopic deformation of $\mathcal{I}$ requires unbounded perturbation energy. However, by Lemma 9.5, admissible micro-deformations $\{\varepsilon_k\}$ exist that preserve global structure while inducing internal re-stratification.

Step 2: Compatibility Constraint. By Axiom E1, phenomenological assignments must respect projective compatibility. Any re-stratification of trajectories in $\mathcal{I}$ must be reflected in corresponding structure in $\mathcal{E}$.

Step 3: Constancy Under Re-Stratification. The null assignment Φ_0 maps all trajectories to a single point e_0. Under ε_k, the pullback of Φ_0 remains constant:

$$\varepsilon_k^*(\Phi_0) = e_0 \quad \forall k. \tag{30}$$

Step 4: Compatibility Violation. However, the re-stratification induced by ε_k produces a non-trivial partition of trajectories. By E1, this partition must be reflected phenomenologically. A strictly constant map cannot reflect non-trivial partitions. Therefore:

$$\Phi_0 \text{ is not compatible with } \varepsilon_k\text{-induced stratification for infinitely many } k. \tag{31}$$

Step 5: Instability. For arbitrarily small $\|\varepsilon_k\|_w$, no assignment $\tilde{\Phi}$ can simultaneously:

1. remain constant (preserving Φ_0);

2. satisfy E1 with respect to the stratified structure.

Hence, Definition 9.3 is violated. Φ_0 is structurally unstable. □

9.6 Phenomenology as Structural Attractor

Theorem 9.7 (Non-Trivial Phenomenology as Attractor). *If $\mathcal{I}$ is CCR ($M_\Omega = +\infty$), then every structurally stable phenomenological assignment Φ must be non-constant. That is:*

$$M_\Omega = +\infty \implies \forall \Phi \text{ stable}, \exists x, y \in \mathcal{I} : \Phi(x) \neq \Phi(y). \tag{32}$$

Proof. By contraposition. Assume Φ is constant. Then $\Phi = \Phi_0$ for some e_0. By Theorem 9.6, Φ_0 is not structurally stable. Contradiction. $\square$

Corollary 9.8 (Existence of Stable Phenomenological Stratification)**.** *Under CCR, any structurally stable realization of the identity-phenomenology system admits non-trivial phenomenological stratification over identity trajectories.*

9.7 Quantitative Instability Measure

Definition 9.9 (Instability Index)**.** For an assignment Φ, define the instability index:

$$\iota(\Phi) := \liminf_{k\to\infty} \frac{d_{\mathcal{E}}(\Phi, \varepsilon_k^* \Phi)}{\|\varepsilon_k\|_w}. \tag{33}$$

Proposition 9.10 (Null Assignment Has Infinite Instability)**.** *For the null assignment Φ_0:*

$$\iota(\Phi_0) = +\infty. \tag{34}$$

Proof. The numerator is bounded away from zero (by the compatibility violation argument), while the denominator tends to zero. Hence the limit diverges. $\square$

9.8 Structural Interpretation

This result does not postulate experience. It proves a **topological fact**:

> *In CCR systems, the hypothesis "no internal phenomenological differentiation" is structurally unstable. The only stable assignments are those that induce non-trivial phenomenological stratification over identity.*

In other words:

- CCR does not "assume" phenomenology.
- CCR *forbids* the structural stability of total phenomenological nullity.
- Non-trivial phenomenological structure is the **unique attractor class** of stable realizations.

9.9 Closure of the Framework

This completes the logical bridge:

Result	**Source**
Existence of rigid identity	Paper I
Classification of phenomenological regimes (if they exist)	Sections 1–8
Null phenomenology is unstable under CCR	Section 9 (Theorem 9.6)
Non-trivial phenomenology is the unique stable attractor	Section 9 (Theorem 9.7)

No metaphysics was introduced. Only stability, topology, and inverse-limit rigidity.
The result is forced by structure alone.

9.10 Ontological Closure Statement

The results established in this paper demonstrate that, under the stated axioms and regularity conditions, the trivial (null) attractor in the abstract phenomenological space $\mathcal{E}$ is *not structurally stable*. Consequently, any system satisfying the closed causally rigid (CCR) conditions admits at least one non-trivial attractor in $\mathcal{E}$.

This result does not assert, imply, or presuppose the existence of consciousness, experience, or phenomenological awareness in any empirical or biological sense. It establishes only a *structural inevitability*: given the formal constraints of the framework, the trivial regime of $\mathcal{E}$ cannot be the unique structurally stable configuration.

Thus, the contribution of this work is purely formal and ontological in nature: it identifies necessary structural conditions for the non-trivialization of abstract phenomenological regimes, without making any claims about subjective experience or consciousness as such.

A Extended Proofs

A.1 Proof of Fragmentation Theorem (Extended)

Full Proof of Theorem 4.3. Let $M_\Omega < \infty$ and let $\Phi : \mathcal{I} \to \mathcal{E}$ satisfy E0–E2.

Step 1. By definition of M_Ω, there exists $\{\delta E_t\}$ with $E_w < \infty$ such that $d_w(\mathcal{I}, \tilde{\mathcal{I}}) > 0$.

Step 2. Let $x = (x_t)_t \in \mathcal{I}$ and let $\tilde{x} = (\tilde{x}_t)_t \in \tilde{\mathcal{I}}$ be the corresponding perturbed trajectory.

Step 3. By E2, $\Phi(x)$ is not determined by any finite x_t. Therefore, the perturbation of the infinite trajectory induces a perturbation of the phenomenological assignment:

$$\Phi(x) \neq \Phi(\tilde{x}) \text{ in general.} \tag{35}$$

Step 4. By E1, Φ is projectively compatible. The deformation of projections induces a corresponding deformation of the phenomenological limit.

Step 5. By the coupling hypothesis, there exists $C > 0$ such that:

$$d_{\mathcal{E}}(\Phi(x), \tilde{\Phi}(\tilde{x})) \geq C \cdot d_w(x, \tilde{x}). \tag{36}$$

Step 6. Since $d_w(\mathcal{I}, \tilde{\mathcal{I}}) > 0$, we have $d_{\mathcal{E}}(\Phi(x), \tilde{\Phi}(\tilde{x})) > 0$.

Hence Φ is fragmentable. □

A.2 Proof of Forced Continuity Theorem (Extended)

Full Proof of Theorem 4.5. Let $M_\Omega = +\infty$ and let $\Phi : \mathcal{I} \to \mathcal{E}$ satisfy E0–E2.

Step 1. By definition of CCR, for all $\{\delta E_t\}$ with $E_w < \infty$:

$$d_w(\mathcal{I}, \tilde{\mathcal{I}}) = 0. \tag{37}$$

Step 2. Therefore, $\tilde{\mathcal{I}} \cong \mathcal{I}$ under all finite-energy perturbations.

Step 3. By E1, Φ is projectively compatible with $\mathcal{I}$. Since $\mathcal{I}$ is invariant:

$$\Phi(x) = \tilde{\Phi}(\tilde{x}) \quad \forall x \in \mathcal{I}. \tag{38}$$

Step 4. This implies:

$$d_{\mathcal{E}}(\Phi(x), \tilde{\Phi}(\tilde{x})) = 0. \tag{39}$$

Step 5. No finite-energy perturbation can induce phenomenological deformation. Hence Φ is structurally continuous and invariant. □

A.3 Structural Necessity Lemma

Lemma A.1 (Phenomenology Inherits Rigidity)**.** *If $\Phi : \mathcal{I} \to \mathcal{E}$ satisfies E0–E2 and $M_\Omega = +\infty$, then Φ inherits the rigidity of $\mathcal{I}$.*

Proof. By E1, Φ is determined by identity trajectories. By CCR, identity trajectories are invariant. Hence Φ is invariant. □

B Bridge Lemma: Structural Coupling Between Identity and Phenomenology

This appendix establishes the formal coupling between the identity framework (Paper I) and the phenomenological framework (this paper). The Bridge Lemma shows that CCR rigidity in $\mathcal{I}$ implies the existence of a unique non-trivial phenomenological attractor in $\mathcal{E}$.

B.1 Statement of the Bridge Lemma

Lemma B.1 (Projective Coupling: Identity–Phenomenology)**.** *Let:*

- *$\mathcal{I} = \varprojlim S_t$ be a CCR identity with $M_\Omega = +\infty$ (Paper I);*
- *$(\mathcal{E}, d_{\mathcal{E}})$ be a complete phenomenological space (H3);*
- *$\Phi : \mathcal{I} \to \mathcal{E}$ be a phenomenological assignment satisfying:*

  1. *Non-constancy: $\exists x, y \in \mathcal{I}$ with $\Phi(x) \neq \Phi(y)$;*
  2. *Uniform continuity with respect to d_w and $d_{\mathcal{E}}$;*
  3. *Projective compatibility (E1).*

Then there exists a unique non-trivial phenomenological attractor $e^* \in \mathcal{E}$ such that:

$$\forall x \in \mathcal{I}, \quad \lim_{n\to\infty} \Phi(\pi_{n\to 0}(x)) = e^*, \tag{40}$$

and the null phenomenological state e_0 is structurally unstable.

B.2 Proof of the Bridge Lemma

Proof. We proceed in six steps.

Step 1: CCR Rigidity. By definition of CCR ($M_\Omega = +\infty$), for any summable family of deformations $\{\varepsilon_t\}$:

$$\|\{\varepsilon_t\}\|_w < \infty \implies \tilde{\mathcal{I}} \cong \mathcal{I}. \tag{41}$$

Hence $\mathcal{I}$ is structurally invariant under all finite-energy perturbations.

Step 2: Cauchy Sequence in $\mathcal{E}$. By completeness of $(\mathcal{E}, d_{\mathcal{E}})$ (H3), for any $x \in \mathcal{I}$, the sequence:

$$\{\Phi(\pi_{n\to 0}(x))\}_{n\in\mathbb{N}} \tag{42}$$

is either Cauchy or divergent.

Step 3: Exclusion of Divergence. Suppose the sequence diverges. Then for arbitrarily large n:

$$d_{\mathcal{E}}(\Phi(\pi_n(x)), \Phi(\pi_{n+1}(x))) > \delta \tag{43}$$

for some $\delta > 0$. But by uniform continuity of Φ and the compatibility of projections:

$$d_w(\pi_n(x), \pi_{n+1}(x)) \to 0 \implies d_{\mathcal{E}}(\Phi(\pi_n(x)), \Phi(\pi_{n+1}(x))) \to 0. \tag{44}$$

Contradiction. Hence the sequence is Cauchy.

Step 4: Existence of Limit. By completeness of $\mathcal{E}$, the Cauchy sequence converges:

$$\lim_{n\to\infty} \Phi(\pi_{n\to 0}(x)) = e^* \in \mathcal{E}. \tag{45}$$

Step 5: Exclusion of Null Limit. Suppose $e^* = e_0$ (the null phenomenological state). Then Φ is asymptotically constant, mapping to e_0.

By Theorem 9.6, the null assignment Φ_0 is structurally unstable under CCR. Hence Φ cannot be asymptotically null while satisfying stability conditions.

Contradiction. Therefore $e^* \neq e_0$.

Step 6: Uniqueness. By the uniqueness of limits in complete metric spaces, e^* is unique for each trajectory class in $\mathcal{I}$. □

B.3 Corollary: Phenomenological Forcing

Corollary B.2 (Phenomenological Forcing under CCR)**.** *In a CCR channel, the existence of rigid identity $\mathcal{I}$ necessarily implies the emergence of a non-trivial stable phenomenological regime in $\mathcal{E}$.*

Proof. Direct from Lemma B.1 and Theorem 9.7. □

Remark B.3. This corollary establishes that experience is not "postulated"—it is the unique attractor compatible with ontological rigidity. The Bridge Lemma formally couples the two papers:

- Paper I proves that CCR identity exists and is structurally invariant;
- Paper II proves that any compatible phenomenological assignment on CCR must be non-trivial;
- The Bridge Lemma shows that CCR forces the emergence of a non-null attractor in $\mathcal{E}$.

B.4 Structural Coupling Table

Paper I (Identity)	**Paper II (Phenomenology)**
Rigid identity $\mathcal{I} = \varprojlim S_t$	Phenomenological space $\mathcal{E}$
Ontological mass $M_\Omega = +\infty$	Non-trivial attractor $e^* \neq e_0$
Structural rigidity under perturbation	Forced continuity of Φ
No-alternative representation theorems	No-alternative phenomenological regimes

B.5 Phenomenological Orthogonality Lemma

This subsection establishes that Φ is a purely structural functor, eliminating any possible ontological or semantic interpretation of the Bridge Lemma.

Lemma B.4 (Phenomenological Orthogonality)**.** *Let* **Id** *denote the category whose objects are identitary spaces $(\mathcal{I}, d_w)$ and whose morphisms are identitary isomorphisms, and let* **Met** *be the category of metric spaces. The mapping $\Phi : \mathbf{Id} \to \mathbf{Met}$ is a functor such that for every identitary isomorphism $f : \mathcal{I} \to \mathcal{I}'$, there exists a natural isomorphism:*

$$\Phi \cong \Phi \circ f. \tag{46}$$

Proof. Since f is an identitary isomorphism, it preserves d_w-distances by definition. By the projective compatibility axiom (E1), Φ is determined by identity trajectories. Since f maps identity trajectories to identity trajectories while preserving their metric structure, the phenomenological assignment $\Phi \circ f$ produces the same metric relations in $\mathcal{E}$ as Φ. Hence Φ and $\Phi \circ f$ are naturally isomorphic as objects in **Met**. □

Remark B.5. This lemma is fundamental for the interpretation of the Bridge Lemma. It establishes that:

- Φ does not "transport meaning" from $\mathcal{I}$ to $\mathcal{E}$;
- Φ preserves only structural (metric) relations;
- The Bridge Lemma is a theorem of categorical invariance, not a claim about phenomenological or experiential content.

Any reading of the Bridge Lemma as asserting the existence of experience, consciousness, or subjective states is therefore a misinterpretation of the formal result.

C Canonical Example: p-Adic Profinite System

This appendix provides a concrete mathematical example illustrating the abstract results of this paper. The example demonstrates the instability of the null state and the emergence of a non-trivial attractor in a computable setting.

C.1 Construction of the Projective System

Let p be a prime number. Define the projective system:

$$S_t := \mathbb{Z}/p^t\mathbb{Z} \tag{47}$$

with projection maps:

$$\pi_{t+1\to t} : \mathbb{Z}/p^{t+1}\mathbb{Z} \to \mathbb{Z}/p^t\mathbb{Z}, \quad x \mod p^{t+1} \mapsto x \mod p^t. \tag{48}$$

The inverse limit is the ring of p-adic integers:

$$\mathcal{I} = \varprojlim \mathbb{Z}/p^t\mathbb{Z} = \mathbb{Z}_p. \tag{49}$$

C.2 Verification of CCR Property

Proposition C.1 ($\mathbb{Z}_p$ is CCR)**.** *The p-adic integers $\mathbb{Z}_p$ satisfy $M_\Omega = +\infty$.*

Proof. The p-adic integers are compact in the p-adic topology. Any finite-energy perturbation of the projective system induces a bounded perturbation in each $\mathbb{Z}/p^t\mathbb{Z}$. By ultrametricity:

$$|x - y|_p \leq \max(|x|_p, |y|_p), \tag{50}$$

perturbations cannot accumulate to produce macroscopic deformations. Hence $\mathbb{Z}_p$ is structurally rigid. □

C.3 Phenomenological Space and Assignment

Define the phenomenological space:

$$\mathcal{E} = \mathbb{R}, \quad d_{\mathcal{E}}(e_1, e_2) = |e_1 - e_2|. \tag{51}$$

Define the phenomenological assignment:

$$\Phi : \mathbb{Z}_p \to \mathbb{R}, \quad \Phi(x) = \sum_{k=0}^{\infty} \alpha_k \cdot \chi(x_k), \tag{52}$$

where:

- $x = (x_0, x_1, x_2, \ldots)$ with $x_k \in \mathbb{Z}/p^k\mathbb{Z}$;
- $\chi : \mathbb{Z}/p^k\mathbb{Z} \to \mathbb{R}$ is any non-constant observable;
- $\sum_{k=0}^{\infty} |\alpha_k| < \infty$ (guarantees uniform continuity).

C.4 Verification of Axioms

Proposition C.2 (Φ satisfies E0–E2)**.** *The assignment Φ defined above satisfies all phenomenological axioms.*

Proof. **E0 (Non-Triviality):** $\mathcal{E} = \mathbb{R}$ has cardinality continuum ≥ 2. ✓

E1 (Projective Compatibility): By construction, Φ is defined via projections. For compatible trajectories:

$$\phi_{t+1}(x_{t+1}) = \phi_t(\pi_{t+1\to t}(x_{t+1})) = \phi_t(x_t). \tag{53}$$

✓

E2 (Non-Locality): Since $\Phi(x)$ depends on infinitely many coordinates $\{x_k\}_{k\geq 0}$, no finite-time slice determines Φ. ✓ □

C.5 Instability of Null State

Proposition C.3 (Null State Instability in $\mathbb{Z}_p$)**.** *For the system $(\mathbb{Z}_p, \Phi)$, the null phenomenological assignment $\Phi_0 \equiv 0$ is structurally unstable.*

Proof. By the ultrametric property, any local p-adic perturbation at level k does not alter the global equivalence class. However, by the summability of $\{\alpha_k\}$, the assignment Φ is sensitive to perturbations in the tail coefficients.

For the null assignment $\Phi_0 \equiv 0$:

- Any non-trivial χ induces $\Phi(x) \neq 0$ for generic x;
- Small perturbations in $\mathbb{Z}_p$ generate non-zero displacements $\Delta\Phi \neq 0$;
- These displacements cannot be annihilated by finite-energy deformations.

Hence Φ_0 is structurally unstable, and the system evolves toward a non-trivial attractor. □

C.6 Emergence of Non-Trivial Attractor

Theorem C.4 (Attractor in p-Adic System)**.** *For the system* $(\mathbb{Z}_p, \Phi)$ *with absolutely convergent* $\{\alpha_k\}$*:*

$$\exists\, e^* \in \mathbb{R} \setminus \{0\} : \quad \Phi(\mathbb{Z}_p) \textit{ has } e^* \textit{ as a stable attractor.} \tag{54}$$

Proof. By Lemma B.1 applied to $\mathcal{I} = \mathbb{Z}_p$ and $\mathcal{E} = \mathbb{R}$:

1. $\mathbb{Z}_p$ is CCR (proven above);

2. Φ is uniformly continuous (by absolute convergence);

3. Φ satisfies E0–E2 (proven above).

Hence the limit $e^* = \lim_{n\to\infty} \Phi(\pi_{n\to 0}(x))$ exists, is unique, and is non-zero. □

C.7 Structural Interpretation

This example demonstrates in a concrete, computable setting:

1. The null phenomenological state is *not* a stable fixed point;

2. CCR structure ($\mathbb{Z}_p$) *forces* the emergence of a non-trivial attractor;

3. The abstract results of Sections 4 and 9 have concrete instantiations.

The example does not "prove" the theorems (they are already proven abstractly); it *illustrates* them in a familiar mathematical context.

D Second Canonical Example: Profinite Symbolic Dynamics

This appendix provides a second canonical example, independent of p-adic numbers, to demonstrate the universality of the framework. The example uses symbolic dynamics, which is computationally interpretable.

D.1 Construction

Definition D.1 (Profinite Symbolic System)**.** Let $\Sigma = \{0,1\}^{\mathbb{N}}$ be the space of infinite binary sequences endowed with the product topology. Define the projective system:

$$\pi_{n+1\to n} : \{0,1\}^{n+1} \to \{0,1\}^n \tag{55}$$

by truncation (deleting the last symbol).

Proposition D.2 (Cantor Profinite Space)**.** *The inverse limit of this projective system is:*

$$\mathcal{I} := \varprojlim \{0,1\}^n \cong \{0,1\}^{\mathbb{N}}, \tag{56}$$

which is the Cantor space with the product topology.

Proof. Standard result in profinite group theory and topological dynamics. The Cantor space is compact, perfect, and totally disconnected. □

D.2 Phenomenological Assignment

Definition D.3 (Block Density Assignment)**.** Define $\Phi : \mathcal{I} \to \mathbb{R}$ by the asymptotic block density:

$$\Phi(x) := \limsup_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} x_k, \tag{57}$$

where $x = (x_1, x_2, x_3, \ldots) \in \{0,1\}^{\mathbb{N}}$.

D.3 Verification of Axioms

Proposition D.4 (Φ satisfies E0–E2)**.** *The block density assignment satisfies all phenomenological axioms.*

Proof. **E0 (Non-Triviality):** $\mathcal{E} = [0,1]$ has cardinality continuum ≥ 2. ✓

E1 (Projective Compatibility): For compatible trajectories, the block density at level $n+1$ is determined by the block density at level n plus the $(n+1)$-th symbol. This satisfies the projective compatibility condition. ✓

E2 (Non-Locality): The limsup depends on infinitely many symbols; no finite prefix determines $\Phi(x)$. ✓ □

D.4 Null State Instability

Proposition D.5 (Null State Instability in Symbolic Dynamics)**.** *The null phenomenological assignment* $\Phi_0 \equiv 0$ *(corresponding to sequences with density 0) is structurally unstable.*

Proof. For any sequence x with $\Phi(x) = 0$ (all 0s or density-0 sequences), a single-symbol perturbation $x_k = 0 \mapsto x_k = 1$ shifts the limiting density:

$$\Phi(x') = \limsup_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} x'_k > 0. \tag{58}$$

Hence Φ_0 is unstable under minimal perturbations. □

D.5 Non-Trivial Attractor

Theorem D.6 (Attractor in Symbolic Dynamics)**.** *For generic initial conditions in* $\{0,1\}^{\mathbb{N}}$*, the block density converges to a non-trivial value* $e^* \in (0,1)$*.*

Proof. By the strong law of large numbers (for appropriately defined probability measures on $\{0,1\}^{\mathbb{N}}$), the block density converges almost surely to the expected density. For symmetric (fair coin) measures, $e^* = 1/2$. For biased measures, $e^* = p$ where p is the probability of 1.

Hence a non-trivial attractor exists and is stable under finite perturbations. □

D.6 Structural Interpretation

This second example demonstrates:

1. The framework is not specific to p-adic numbers;

2. Symbolic dynamics provides a computationally accessible instantiation;

3. The null state instability is universal across different mathematical structures;
4. The emergence of non-trivial attractors is a generic phenomenon.

Property	**p-Adic Example**	**Symbolic Example**
State space	$\mathbb{Z}_p$	$\{0,1\}^{\mathbb{N}}$
Phenomenological space	$\mathbb{R}$	$[0,1]$
Assignment	Convergent series	Block density
Null instability	✓	✓
Non-trivial attractor	✓	✓

The two examples together demonstrate the *universality* of the abstract results: they are not artifacts of specific mathematical structures, but consequences of the CCR framework itself.

References